

Behaviour

A textbook on Behaviour

by Alison Wilson

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Contents

Behavioural Science

Definition and Scope of Behavioural Science

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Behavioural science is the systematic, evidence-based study of what organisms do — their actions, reactions, and interactions with their environments and with one another — together with the processes that generate, sustain, and alter those actions. This subsection establishes what the term denotes, how the field is demarcated from neighbouring disciplines, and what falls inside and outside its scope. Later subsections examine the history, biological bases, cognitive processes, learning mechanisms, and social influences that the field studies; the purpose here is to fix the object of study and the boundaries of the enterprise.

Defining Behaviour as the Object of Study

Any science is defined in the first instance by its dependent variable. For behavioural science, that variable is *behaviour*: activity of an organism that is in principle observable, measurable, and extended in time. A rat pressing a lever, a shopper abandoning an online basket, a parliament voting on a carbon tax, and a flock of starlings turning as one are all behaviours in this sense — they are events, not states of mind, and they can be counted, timed, coded, or otherwise operationalised.

Two clarifications matter.

Behaviour is not the same as mind. Behavioural science has historically been accused of denying inner life, and some early programmes did restrict themselves to observable responses (a position examined in the Foundations of Behavioural Psychology subsection). Contemporary behavioural science is not eliminativist: it treats thoughts, feelings, and intentions as legitimate causes, provided they are operationalised well enough to yield measurable predictions. Self-report scales, reaction times, physiological indices, and choice data are all admissible, because all are behavioural measures in the broad sense.

Behaviour is not the same as behaviourism. *Behaviourism* is one theoretical tradition within behavioural science — a family of positions about how behaviour should be explained. *Behavioural science* is the wider interdisciplinary field. A cognitive neuroscientist imaging reward anticipation and a sociologist coding protest repertoires are both behavioural scientists, and neither is a behaviourist.

Behaviour is also classified along several dimensions that recur throughout this book:

Dimension	Contrast	Example
Observability	overt vs. covert	Walking vs. privately rehearsing an argument
Level	individual vs. dyadic vs. collective	One person voting vs. a crowd dispersing
Timescale	phasic vs. tonic vs. developmental	A startle reflex vs. a habit vs. a career
Function	instrumental vs. expressive	Working for pay vs. laughing at a joke

What Makes the Study of Behaviour a Science?

Behavioural science earns the label "science" by meeting the same demands as any empirical discipline: publicly specified constructs, systematic observation, quantification, replicability, and theories that generate falsifiable predictions. Its distinctive methodological problems — the reactive subject, the difficulty of randomising life histories, the sheer multiplicity of causes — are treated in Research Methods and Applied Behavioural Science.

A useful way to describe the ambitions of the field is the threefold scheme inherited from experimental analysis of behaviour: **description** (what happens, under what conditions, at what rate), **prediction** (what will happen next, given the current state and context), and **control or explanation** (what interventions change it, and why). Each tier presupposes the one before it. This ordering disciplines the field's claims: a model that predicts a choice but cannot be perturbed to change it remains scientifically incomplete.

flowchart LR

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A[Observation and description] --> B[Prediction]
B --> C[Causal explanation]
C --> D[Intervention]
D --> A
```

The Disciplinary Composition of Behavioural Science

The field is genuinely interdisciplinary, and the plural form "behavioural sciences" is often used to make this explicit. Its principal contributing traditions include:

- **Psychology** — experimental, cognitive, developmental, social, clinical, and comparative.
- **Behavioural biology and ethology** — the adaptive function and phylogenetic history of behaviour, including in non-human species.
- **Neuroscience** — the neural mechanisms that implement behaviour.
- **Behavioural economics** — decision-making under bounded rationality, treated in the Behavioural Economics part of this book.
- **Sociology, anthropology, and linguistics** — norms, institutions, culture, and communicative practice.
- **Epidemiology, public health, education, criminology, and computer science** — applied and computational branches, including machine learning of behavioural data.

These traditions converge on the same dependent variable while differing in method, timescale, and explanatory vocabulary.

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flowchart LR
    P[Psychology] --> BS[Behavioural science]
    B[Biology and ethology] --> BS
    N[Neuroscience] --> BS
    E[Economics] --> BS
    S[Sociology and anthropology] --> BS
    C[Computer science] --> BS
    BS --> O[Explanation, prediction, intervention]

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Levels of Analysis

The single most useful organising idea in the field is that behaviour can be described and explained at multiple levels, none of which is inherently privileged. A standard ascending sequence runs from the molecular and neural, through the cognitive and individual, to the dyadic, group, institutional, and cultural. Explanation at one level does not compete with explanation at another; it answers a different question.

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flowchart TD
    A[Neural and physiological] --> B[Cognitive and affective]
    B --> C[Individual behaviour]
    C --> D[Dyadic and small group]
    D --> E[Institutional and organisational]
    E --> F[Cultural and population]

```

The classic integrative framework for managing these levels is Tinbergen's four questions, which hold that any behaviour is fully explained only when four *separate* questions are answered.

Question	Asks about	Example: prosocial helping
Mechanism	Immediate causal machinery	Empathy circuits, arousal, decision processes
Ontogeny	Development over the lifespan	Parental modelling, socialisation into reciprocity
Function	Adaptive significance	Reputation, reciprocity, group cohesion
Phylogeny	Evolutionary history	Comparative evidence in other primates

Conflating these questions is a common source of sterile debate — for instance, arguing that a neural account of altruism "refutes" an evolutionary one, when the two are answers to different questions.

Worked example: an impulse purchase at the checkout

Consider a shopper who adds a chocolate bar to a grocery order at the last moment. Behavioural science can describe this event at several levels simultaneously:

- **Neural:** cue-triggered dopamine anticipation as the display enters the visual field.
- **Cognitive:** the discounted price serves as an anchor; the item's availability makes it salient.
- **Learning:** previous purchases were reinforced by immediate consumption, building a strong cue-response association.
- **Social:** perceived norms ("everyone buys a little treat"), household roles, and gift-giving conventions.

- **Institutional:** placement at the point of sale, frictionless payment, and credit availability are design choices made by others.
- **Economic:** time-inconsistent preferences and present bias, the subject of the chapter on Present Bias and Self-Control.

No single level is the "real" explanation. A complete account is a coordinated set of answers, and an intervention may target any of them — moving the display, changing the default, or teaching self-regulation.

Core Assumptions and Commitments

Despite their diversity, behavioural scientists share a recognisable set of commitments:

1. **Determinism, weakly stated.** Behaviour is treated as lawful and predictable in principle, even where prediction is practically imperfect.
2. **Empiricism.** Claims are adjudicated by observation, not by intuition or authority alone.
3. **Operationalism.** Abstract constructs must be tied to measurable indicators, or they cannot be tested.
4. **Parsimony.** Simpler explanations are preferred until the data demand otherwise.
5. **Interactionism.** Behaviour is a joint product of person and situation; neither trait nor context alone suffices.
6. **Cumulative theory.** Findings are meant to refine general models, not merely accumulate curiosities.

Contemporary variants add commitments to computational modelling, to dual-process accounts of judgement, and increasingly to prediction and control in real-world settings rather than the laboratory alone.

Scope and Boundaries

The scope of behavioural science can be mapped on three axes: **level of analysis** (neural to cultural), **timescale** (milliseconds to evolutionary), and **domain of application** (health, finance, education, law, environment, technology). The field deliberately spans all three. Boundary questions are nevertheless live:

- **Is subjective experience in scope?** Yes, insofar as it can be measured and shown to do explanatory work — but it is not the field's defining object.
- **Are institutions in scope?** Yes, as causes and consequences of individual behaviour, and as settings where behaviour is aggregated. The sociology of institutions is thus adjacent rather than foreign.
- **Are non-human animals in scope?** Yes; comparative work is foundational to learning theory and to behavioural ecology.
- **Are artificial systems in scope?** Increasingly, where their behaviour is studied with the field's methods, or where they are used as models and as interventions.

By contrast, the field does not claim to settle normative questions. Behavioural science can establish that a nudge changes saving rates; it cannot establish that higher saving rates are desirable. Nor does it subsume the content of economics, law, or ethics — it informs them.

A Brief Note on Applied Scope

The applied reach of behavioural science has expanded sharply since the 2000s through behavioural insights units in governments, the growth of behavioural finance and marketing, the design of digital choice

architectures, and public health campaigns. These applications are treated in Behavioural Economics in Policy and Markets and in Behaviour Modification and Applied Extensions. Their prominence has also sharpened ethical scrutiny: interventions that shape behaviour without notice raise questions of autonomy, transparency, and consent, and the field increasingly treats these as part of its subject matter rather than as external constraints.

Limits and Open Questions

Three critical issues define the current frontier of scope. First, **generalisability**: much of the evidence base derives from participants and settings that are unrepresentative of humanity as a whole, which constrains how far findings extend. Second, **replication and effect size**: several headline results in social and behavioural research have proved fragile, prompting methodological reform rather than abandoning the enterprise. Third, **integration**: the field's very breadth makes cumulative theory difficult, and the persistent challenge is to connect levels of explanation without collapsing them into one another. Whether behavioural science is best understood as a single discipline or as a federation of disciplines sharing a dependent variable remains an open question — arguably the field's defining tension.

Summary

- Behavioural science is the systematic study of what organisms do, together with the processes that produce and change it.
- Its object is behaviour: observable, measurable, temporally extended activity. It is not reducible to introspection, and it is not the same thing as behaviourism.
- Its ambitions run from description to prediction to causal explanation and intervention, each tier presupposing the last.
- It is interdisciplinary by constitution, drawing on psychology, biology, neuroscience, economics, sociology, anthropology, and computational fields.
- Levels of analysis run from neural to cultural; Tinbergen's four questions (mechanism, ontogeny, function, phylogeny) discipline how they are combined.
- The field is unified by shared methodological commitments — empirical, operational, parsimonious, interactionist — rather than by a single theory.
- Its boundaries are contested at the edges (institutions, experience, artificial systems) and its scope excludes normative judgement, which it informs but cannot supply.